Refined P_α Overlap: Parametric Spectroscopic Factor and Self-Consistent WS+Coulomb Eigenmode

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Abstract

Two comparison channels extend the throat+Woods–Saxon overlap proxy [3] on the same 14-isotope set. **Channel A** extracts per-isotope $S_i = P_{\alpha,\mathrm{ext}}/P_{\mathrm{model},i}$ (diagnostic), retains one global S, and fits predictive parametric $\log_{10} S(A,Z)$ and $\log_{10} S(R)$ with leave-one-out (LOO) RMS vs extracted $\log_{10} P$. **Channel B** replaces the phenomenological WS trial by a self-consistent $\ell = 0$ eigenmode of $V_{\mathrm{WS}} + V_C$ on $[0,R]$ (Dirichlet; depth V_0 so $E_0 = Q_\alpha$), overlapped with the throat-weighted flag mode. On the stated set, parametric $S(A,Z)$ LOO RMS = 0.2862 reduces residuals most among *non-tautological* protocols (vs throat+WS global-S RMS = 0.8247). The eigenmode raises Pearson r to $+0.9676$ but leaves global-S RMS essentially unchanged (0.8257). RC1 scope: documented proxies — not uniqueness.

1 Motivation

Reference [3] closed flat→throat/WS geometry and listed isotope-resolved S and self-consistent WS+Coulomb eigenmodes as open. This note implements both as comparison channels on the unchanged table of [1].

Remark (RC1 scope). Per-isotope S_i zeros that isotope's residual by construction and is reported only as a diagnostic. Predictive claims use parametric S with LOO, or Channel B with one global S. No uniqueness of many-body preformation is claimed.

2 Channel A — isotope S and parametric models

Base model: throat+WS of [3] ($w = R/r$, Woods–Saxon trial). Define

$$S_i = \frac{P^{(i)}_{\alpha,\mathrm{ext}}}{P_{\mathrm{model},i}} = 10^{\log_{10} P^{(i)}_{\mathrm{ext}} - \log_{10} P_{\mathrm{model},i}}. \tag{1}$$

Global $S_\star = 10^{\langle \log_{10} P_{\mathrm{ext}} - \log_{10} P_{\mathrm{model}} \rangle}$ as before. Parametric fits (least squares):

$$\log_{10} S = b_0 + b_1 A + b_2 Z, \tag{2}$$

$$\log_{10} S = c_0 + c_1 R. \tag{3}$$

Leave-one-out: for each isotope, refit on the other 13 and predict $\log_{10} S$; LOO RMS is on predicted $\log_{10}(S P_{\mathrm{model}})$ vs extracted $\log_{10} P$.

2.1 Channel A numbers (stated set)

$$\langle S_i \rangle = 1.001 \times 10^{-1} \quad (\sigma = 1.260 \times 10^{-1}),$$

$$\langle \log_{10} S_i \rangle = -1.4979 \quad (\sigma = 0.8558),$$

$$S_\star = 3.178 \times 10^{-2}, \qquad \mathrm{RMS}_{S_\star} = 0.8247,$$

$$\mathrm{LOO\,RMS}_{S(A,Z)} = 0.2862, \qquad \mathrm{LOO\,RMS}_{S(R)} = 0.4695.$$

Fit (in-sample): $b_0 \approx 14.616$, $b_1 \approx +0.0979$, $b_2 \approx -0.4312$; $c_0 \approx 48.503$, $c_1 \approx -5.435\,\mathrm{fm}^{-1}$.

3 Channel B — self-consistent WS+Coulomb eigenmode

For parent (A, Z) with daughter $(A_d, Z_d) = (A-4, Z-2)$, reduced mass μ, and nuclear radius R from the table, solve

$$-\frac{\hbar^2}{2\mu}\,u''(r) + \big[V_{\mathrm{WS}}(r; V_0) + V_C(r)\big]u(r) = E\,u(r) \tag{4}$$

on $[0, R]$ with Dirichlet $u(0) = u(R) = 0$ (finite-difference Sturm–Liouville). Potentials:

$$V_{\mathrm{WS}} = -\frac{V_0}{1+\mathrm{e}^{(r-R_{\mathrm{WS}})/a}}, \quad R_{\mathrm{WS}} = r_0 A_d^{1/3},\ a = 0.55\,\mathrm{fm}, \tag{5}$$

$$V_C = \begin{cases} \dfrac{Z_\alpha Z_d e^2}{2R_C}\big(3-(r/R_C)^2\big), & r < R_C, \\ Z_\alpha Z_d e^2/r, & r \ge R_C, \end{cases} \qquad R_C = R_{\mathrm{WS}}. \tag{6}$$

Self-consistency: adjust V_0 so the ground eigenvalue $E_0 = Q_\alpha$ (box-resonance surrogate; not a complex Gamow eigenphase). Overlap with $u_{\mathrm{in}} = r\, j_0(\pi r/R)$ uses throat weight $w = R/r$ as in [3].

3.1 Channel B numbers (stated set)

$$\begin{aligned} \langle \log_{10} P_{\mathrm{model}} \rangle &= -0.1110, \\ r\big(\log_{10} P_{\mathrm{model}}, \log_{10} P_{\mathrm{ext}}\big) &= +0.9676, \\ \mathrm{RMS}_{S=1} &= 2.0237, \qquad S_\star = 1.421\times 10^{-2}, \qquad \mathrm{RMS}_{S_\star} = 0.8257. \end{aligned}$$

Typical fitted depths: $V_0 \sim 36$–$43\,\mathrm{MeV}$ on this set (shallow relative to deep cluster wells — a consequence of forcing $E_0 = Q_\alpha > 0$ under hard-wall Dirichlet at R).

Remark (Boundary condition). Dirichlet at the throat wall R matches the overlap support of the confined flag mode. Channel C implements outgoing Coulomb/Gamow matching on $[0, b]$; on the stated 14-isotope set it does not reduce global-S RMS vs Channel B (see audit log).

4 Acceptance comparison

Table 1: Flat / throat+WS / refined eigenmode / parametric S on the stated 14-isotope set. LOO applies only to parametric S.

Channel	$\langle \log_{10} P \rangle$	r	$\mathrm{RMS}_{\mathrm{raw}}$	$\mathrm{RMS}_{S_\star}$ / LOO
flat Gaussian	−0.3905	+0.8882	1.7705	0.8220 / —
throat + WS	−0.4607	+0.8875	1.7099	0.8247 / —
refined eigenmode	−0.1110	+0.9676	2.0237	0.8257 / —
Gamow outgoing	−0.1110	+0.9676	2.0237	0.8257 / —
parametric $S(A, Z)$ (LOO)	(base throat+WS)	—	—	LOO 0.2862
parametric $S(R)$ (LOO)	(base throat+WS)	—	—	LOO 0.4695

Remark (Which channel reduces residuals most?). Excluding tautological per-isotope S_i, **Channel A parametric** $S(A, Z)$ yields the largest residual reduction (LOO RMS 0.2862 vs global-S RMS ≈ 0.82). Channel B improves correlation ($\Delta r \approx +0.080$) but does not reduce global-S RMS relative to throat+WS on this set ($\Delta\mathrm{RMS}_{S_\star} \approx +0.001$).

5 Artifacts and protocol

Implementation: `rh_papers_may21/acs-framework/code/palpha_overlap/palpha_overlap_refined.py`. JSON: `docs/palpha_overlap_refined_results.json`. Logs: `docs/palpha_overlap_refined_logs/`. Baselines retained: `palpha_overlap.py`, `palpha_overlap_throat.py`. Gamow channel: `palpha_overlap_gamow.py`. Extended catalog ($n = 29$): `palpha_overlap_extended.py`, `isotope_catalog.py`, `docs/palpha_overlap_extende`
Audit: `docs/palpha_audit_log.md`.

6 Channel C — Gamow outgoing boundary on $[0, b]$

Channel B used Dirichlet $u(0) = u(R) = 0$ on the throat support. Channel C extends the radial domain to $[0, b]$ (b = outer Coulomb turning point) and replaces the hard wall at R with a Robin outgoing match at $r = b$ from WKB slope in the exterior allowed region. Overlap support remains $[0, R]$ with throat weight $w = R/r$.

On the stated 14-isotope set: $r = +0.9676$, $\mathrm{RMS}_{S_\star} = 0.8257$ (vs Dirichlet eigenmode +0.9676, 0.8257). Gamow outgoing BC does not improve global-S RMS under the documented $E_0 = Q_\alpha$ protocol.

7 Extended isotope catalog ($n = 29$)

Fifteen additional alpha emitters from NNDC/NuDat tabulations were added (see extended JSON metadata). On $n = 29$ throat+WS: LOO $\mathrm{RMS}_{S(A,Z)} = 1.355$ vs 0.286 on the original 14; coefficient signs ($b_1 > 0$, $b_2 < 0$, $c_1 < 0$) hold but magnitudes shift. Parametric $S(A, Z)$ predictive power is scoped to the original stated set under the mixed extraction protocol in the audit log.

8 Conclusion

On the stated set, predictive parametric $S(A, Z)$ with leave-one-out compresses residual RMS from ~ 0.82 (one global S) to 0.286. A self-consistent WS+Coulomb interior eigenmode improves correlation with extracted $\log_{10} P$ but does not, by itself, beat throat+WS on global-S RMS under the documented Dirichlet/$E_0 = Q_\alpha$ protocol. Channel C (Gamow outgoing on $[0, b]$) likewise leaves global-S RMS unchanged on the stated set. Extended catalog ($n = 29$) degrades LOO parametric RMS as documented. Absolute uniqueness is not claimed.

References

[1] ACS Theoretical Physics Working Group, *Nuclear Decay as a Spectral Bifurcation of the Flag Condensate*, ACS Framework note (2026), `flag_condensate_nuclear_decay.tex`.

[2] ACS Theoretical Physics Working Group, *Standing-Wave Overlap Proxy for Alpha Preformation*, ACS Framework note (2026), `flag_condensate_palpha_overlap.tex`.

[3] ACS Theoretical Physics Working Group, *Throat-Geometry Overlap for Alpha Preformation*, ACS Framework note (2026), `flag_condensate_palpha_throat_overlap.tex`.